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Deep Learning-based Integrated Framework for stock price movement prediction

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ABSTRACT

Stock market prediction is a very important problem in the economics field. With the development of machine learning, more and more algorithms are applied in the stock market to predict the stock price movement. However, stock market prediction is regarded as a challenging task for the noise and volatility of stock market data. Therefore, in this paper, a novel hybrid model SA-DLSTM is proposed to predict stock market and simulation trading by combine a emotion enhanced convolutional neural network (ECNN), the denoising autoencoder (DAE) models, and long short-term memory model (LSTM). Firstly, user-generated comments on Internet were used as a complement to stock market data, and ECNN was applied to extract the sentiment representation. Secondly, we extract the key features of stock market data by DAE, which can improve the prediction accuracy. Thirdly, we take the timeliness of emotion on stock market into consideration and construct more reliable and realistic sentiment indexes. Finally, the key features of stock data and sentiment indexes are fed into LSTM to make stock market prediction. Experiment results show that the prediction accuracy of SA-DLSTM are superior to other compared models. Meanwhile, SA-DLSTM has a good performance both in return and risk. It can help investors make wise decisions.

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1. Introduction

Stock market prediction has attracted considerable research interests from both theoretical and application perspective. In particular, accurate prediction helps investors seek to maximize profits in the stock market. However, predicting stock market trend is a difficult task for the financial data have the characteristics of non-stationary, uncertainty, and a lot of noise. The generation of data is not only a result of historical market behaviors, but also affected by information such as companies' financial reports [1,2], financial news [3,4], macroeconomy [5] and investors' emotions [6,7]. The richness of data sources can create a more comprehensive understanding of the stock market and make a more accuracy prediction of the stock price than before. Some methods can find the relationship between the historical data and future movement of the stock price, and use historical data to predict the future movement [8]. One of the crucial parts of the prediction method is the technical analysis [9] that can extract features from the original data. Features are subjectively considered that the future movements of the stock price can be influenced by the historical behaviors. Apart from these traditional methods, machine learning methods are also used to learn the relationship between the historical data and future movement of the stock price for they have the strong capability of learning. Zhang et al. applied BP algorithm to predict the stock price patterns [10]. Based on machine learning and mean–variance model, a novel portfolio construction approach was proposed [11]. However, owing to the above methods are shallow learning methods, their accuracy is low.

Among deep learning methods, neural networks play crucial role to extract features from original data. The feature extraction can be performed by designing an integrated network structure that can extract features for special data. Although some models have been designed to learn the trading rule, in the task of extracting feature for predicting price trend, deep learning methods are barely used to design a specialized network structure that adequately considers characteristics of objective tasks. Therefore, Long et al. applied the public market information and transaction records to propose a novel model to predict stock price trend [12]. A predictive model which combined the Convolution Neural Network and decision tree was proposed to analyze the events provided by Times and Trades information [13]. Although the above modules have improved the accuracy of prediction, sequence feature extraction has not been intensively studied.

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The behavioral finance emphasizes that the behavior and emotional factors have played an important role in financial decision-making. Deng et al. used the individual, institutional, and foreign investor sentiment to forecast the movement direction of the Shanghai Stock Exchange index [14]. Based on the social media data that is a valuable resource, the investors' decisions and patterns can be recognized. The investor's emotion becomes an important factor of the financial market [15]. With the rapid development of the Internet, and especially social media, the textual content on the Internet can reflect the investor's sentiment and predict stock market trend. Therefore, an effective method is required to extract knowledge from a large number of textual documents [16]. Sentiment analysis (SA), opinion mining, natural language processing (NLP) and data mining have been used to distill views, attitudes, and emotions from the textual contents [17]. Therefore, this paper utilizes SA to construct the sentiment indexes, and combines the indexes with stock market data to forecast stock movement direction.

Based on the variety of information including the historical financial data and textual contents posted on stock message boards, the investor can make an efficient investment decision. There is a popular supposition that investor's sentiment on social media has impact on the stock markets [18]. The positive sentiment might prompt investors to buy stocks, while negative sentiment may motivate investors to sell shares. The sentiment can drive the fluctuations of the stock market. It is natural for researchers to effectively utilize the sentiment to make an accuracy prediction. Thus, predicting stock price by SA is an essential issue worth studying. The quantization method for sentiment has been paid greatly attentions by many researchers. Hasan et al. used sentiment analysis and simple fuzzy decision to enhance decision [19]. Several approaches take explicit variables as features by making simple calculation on the message board [20]. However, these variables are weak in representing the investor's sentiment since their real opinions are ignored in these variables.

Individuals share their opinions and mood states through social media. The analysts can obtain valuable information by analyzing the textual contents and users' comments on social media. Translating sentiment into number can generate different interpretations of sentiment expression on social media. Hence, detecting sentiment of stock messages is a challenging task. Some literatures [21] use simple methods to take words which are of certain sentiments to compute the emotional index of textual contents. Some researchers consider the sentiment extraction as a classification task, so they utilize machine learning methods to extract sentiment from messages [22]. Ren et al. integrated SA into a support vector machine (SVM) to predict the stock market movement [23]. Weng et al. developed a financial expert system that can calculate the sentiment scores of published news articles for a given stock to predict short term stock prices [24]. According to the above studies, the features applied in the information representation of textual contexts have great impact on the performance.

In this paper, we propose a novel model named SA-DLSTM for feature extraction on the financial time series data and price movement prediction task. Based on feature transformation on sequential market data, data can be mapped into the feature space where SA-DLSTM can discriminate different patterns for classification-based prediction. In SA-DLSTM the denoising autoencoder (DAE) models, SA model and LSTM are combined as an integral network, by which features transformed from different models are merged and information from different views can be obtained. SA model can extract sentiment information from stock messages, and transform the sentiment information into sentiment index. SA-DLSTM takes the sentiment index as the representative dimension of message emotion. SA-DLSTM can integrate the stock-related messages extracted from the news with the investor's sentiment on social media to obtain the interrelations among different information sources. It can calculate the sentiment index from different information sources. Furthermore, SA-DLSTM uses DAE models to extract features from financial market data and technical indicators. DAE can effectively remove noise from data, so there is a very strong correlation between the features extracted by DAE and the original data. These features and sentiment indexes are used as the inputs for LSTM to learn the relationships among the sentiment indexes and the stock market indexes. From the output from LSTM, we can predict the stock market movement direction. SA-DLSTM can take full advantages of different methods, and get rid of the dependence on advanced signal processing techniques and manual feature extraction. We use SA-DLSTM to extract features from high-frequency market data of the Hang Seng Index (HSI), and then recognize the extreme market to exploit the suitable trading signal. A simple prediction-based trading strategy is tested based on signals from SA-DLSTM to evaluate profitability and stability of SA-DLSTM.

The main contributions of this paper are as follows:

(1) Public opinions and sentiments are associated with the actual volatility of stock prices. The combination of various heterogeneous information resources, SA-DLSTM can provide robust and accurate predictions for the stock market trends.
(2) SA-DLSTM transforms textual contents into sentiment indexes in which the indexes are with different emotion strength. In this way, we can accurately estimate the fine-grained sentiment in textual contents. And we use an exponential time function to calculate the timeliness of emotion. Hence, this paper takes the timeliness of emotion on stock market into consideration and constructs more reliable and realistic sentiment indexes.
(3) A novel network based on deep learning methods is proposed for extracting feature from multivariate financial time series data and classification-based prediction. SA-DLSTM can extract features that contain different types of information and integrate features for further classification.

The rest of the paper is organized as follows. Section 2 contains the literature review. Section 3 contains the proposed method. Section 4 discusses the experimental results. Section 5 contains the conclusions and policy implications.

2. Related work

2.1. Sentiment analysis in stock market prediction

SA is a approach to extract views, attitudes, and emotions from textural contents on Internet by NLP, data mining, and computational linguistics. Now many investors express their views about the stock market on social media, so the sentiments on the messages have been considered as a indicator that can reflect the status of stock market. Therefore, the research on SA of stock social media has been promoted by stock market prediction. The relation between investors' sentiments and stock market returns is a key part in behavioral financial [25]. The economists have paid much attention to the investors' sentiments about financial market lately [26]. The empirical literature examined that both investor sentiment and investor behavior have significant impacts on stock returns [27]. Nardo et al. predicted the financial movements using online information from twitter [28]. Tetlock found that the pessimism had a negative impact on the stock market [29]. Gillam et al. quantified the information contained in textual data and discovered that it enhanced earning prediction [30].

Many researchers used different classifier to deal with different problems since the performance of classifiers varies greatly for different objects. SVM, Naive Bayesian classifier (NBC), and dynamic panel data model are used to build a box-office prediction with word-mouth [31]. Wu et al. proposed a approach that combined SA and SVM to predict financial risk measured in volatility terms [22]. Ali et al. proposed a model-based opinion mining by introducing a part-of-speech graphical model to extract investor's opinions [32]. There are many papers studying the correlations between stock price and sentiment, but few studies have studied the role of sentiment analysis for stock prediction, which is exactly what we focus on in this paper.

2.2. Deep learning for time series data

Deep learning methods are utilized in financial time series modeling and prediction for their good capability of learning potential relationships between patterns in features and labels [33]. Chakole et al. used the reinforcing learning to find an optimal dynamic trading strategy by interacting with the actual stock market as its environment [34]. Pinheiro et al. combined character-level language model with recurrent neural networks to model the changes in the stock price [35]. Chen et al. used gold prices, a gold volatility index, crude oil price, and a crude oil price volatility indexes as input features of deep learning model to predict the future stock price trends [36]. Baek et al. proposed a novel framework, which consists of two modules: an overfitting prevention LSTM module and a prediction LSTM module, to predict the stock market index [37]. Lee et al. proposed a new stock market forecasting framework which was an end-to-end integrated neural network framework consisting of two feature extractor modules to successfully learn high-level features from super-high dimensional time-series data [38].

However, most of these studies use the single type network to make prediction of stock price. Although, some of them show perfect performance, the prediction ability of integrated network structure has been barely studied on the financial time series data. The motivation of SA-DLSTM is to explore the performance of integrated network structure based feature extraction and SA on financial time series data.

3. Architectural design of SA-DLSTM based on sentiment index

Based on SA, SA-DLSTM extracts features from social media, news on Internet and stock forum text to automatically obtain the sentiment indexes of other textual contents. We aggregate textual contents for each stock on a daily basis. Therefore, SA-DLSTM model is constructed by SA, DAE models and LSTM to predict future stock price volatility.

3.1. Strategy structure

In this paper, a trading strategy is designed according to SA-DLSTM. For a given trading date t, we can obtain the trading strategy of one stock by:

$$Td(t) = \mathcal{L}\left(\frac{CP_t - CP_{t-1}}{CP_t}\right), \tag{1}$$

where CP_t is the close price on the trading day t. The $\mathcal{L}$ is defined as follows:

$$\mathcal{L}(x) = \begin{cases} -1 & x \le -0.03 \\ 0 & |x| < 0.03 \\ 1 & x \ge 0.03. \end{cases} \tag{2}$$

To obtain the share price moving trends, the movement scope threshold is set as 3%. We consider the $Td(t)$ as up (down) if the price change ratio of the stock is larger than 3% (smaller than 3%) on a trading day. Therefore, a trading signal is sent out on the trading day t. Otherwise, the price is considered as little fluctuation, and the trading signal is not sent out on trading day t.

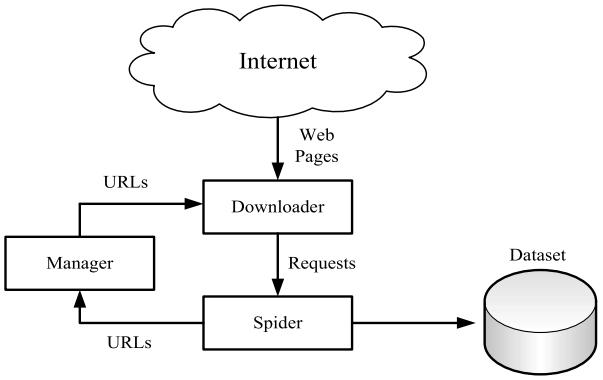


Fig. 1. Architecture of the web crawler.

3.2. Sentiment analysis

Stock market is affected by trading volume, Web news and the investors' emotions. Hence, we must depend on multi-source data to make accurate prediction for stock market trends. Therefore, this section introduces how to gather messages from news and social media data, and obtain sentiment index from them. which is used as the key factor to make accurate prediction.

3.2.1. Web crawler

In this step, we build a web crawler system that has a good performance and can automatically collect information accurately. The crawler system is presented in Fig. 1. The manager node not only managers the list of URLs, but also decides the priority of URL. Then it gives the list of URLs to the downloader. Based on the list, the downloader acquires the web pages and provide them to the spider. The spider can parse the pages and extract the targeted contents. The targeted content includes two parts: one is the textual content with the precious date from the website; the other is the URLs contained in the targeted page. These steps can be repeated until the crawler system obtains all the targeted textual contents. These textual documents are stored in the dataset.

3.2.2. Sentiment analysis

In this paper, we use the sentence-based SA method to process the textual contents. A sentence is considered as a unit to interpret the meaning of the comment since a sentence can express a relatively complete meaning and deal with the ambiguity. Therefore, a whole comment is divided into several sentences, and we use Jieba which is an open-source python component for Chinese text segmentation to segment each sentence. To obtain the sentiment indexes from the textual contents on Internet, we use the emotion enhanced convolutional neural network (ECNN) [17] to extract the sentiment representation. The output of ECNN is the sentiment index that is used as the latent features of the sentiments on social media.

3.2.3. Revised sentiment

It is well known that the large amount of news and posts seems to spring up almost on Internet daily. With a large number of valuable information, investors are very likely to take actions

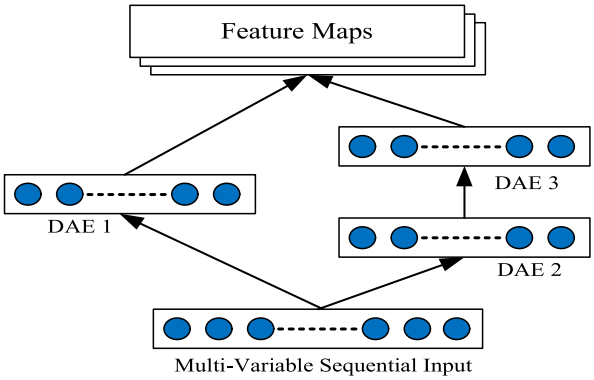


Fig. 2. Multi-Layers for feature extraction on financial time series.

on the next day. In this paper, we evaluate the effect of information by using an exponential time function since information has significant influence if it is more recent. According to [39], we consider the sentiment of everyday as a weighted average of past sentiment on the stock market where the weights decrease exponentially. The function is determined as the following equation:

$$S_t^{rev} = e^{-5\lambda}S_{t-5} + e^{-4\lambda}S_{t-4} + e^{-3\lambda}S_{t-3} + e^{-2\lambda}S_{t-2} + e^{-\lambda}S_{t-1} + S_t, \quad (3)$$

where S_t^{rev} is the revised sentiment index on day t; S_{t-5}, S_{t-4}, S_{t-3}, S_{t-2}, and S_{t-1} represent the sentiment indexes of last 5 days; S_t is the sentiment index of day t; $\lambda(\lambda > 0)$ is prescribed.

3.3. Multi-layers for feature extraction on financial time series

To extract features from original data, a network module is designed by integrating a set of DAE models. Fig. 2 demonstrates that SA-DLSTM integrates both single layer and multi-layers DAE models to obtain discrete information extracted through time series data.

DAE [40] is an extension version of the classical autoencoder (AE). DAE reconstructs the input data by adding random noise, which prevents AE simply learning an identity mapping between the input and the reconstructed output. It can capture the effective features and get the powerful representations from the corrupted input. DAE has an encoder and a decoder. The input $\overrightarrow{\mathrm{x}}$ is corrupted into $\tilde{\mathrm{x}}$ under a certain probability ξ_{noise} by means of a stochastic mapping $\tilde{\mathrm{x}} \sim D(\tilde{\mathrm{x}}|\overrightarrow{\mathrm{x}}, \xi_{noise})$. D is a type of distribution that is determined by the raw distribution of the input data and the type of random noise added to the input data.

Encoder: Based on a nonlinear activation function $f(\cdot)$, the encoder maps the input from a high-dimensional space into a low-dimensional space. The encoder converts $\overrightarrow{\mathrm{x}}((x_1, x_2, \ldots, x_n)(x_i \in \mathbb{R}^n))$ into a hidden representation $\overrightarrow{\mathrm{y}}((y_1, y_2, \ldots, y_m)(y_i \in \mathbb{R}^m)(n > m))$. The typical form is $\overrightarrow{\mathrm{y}} = f(W\overrightarrow{\mathrm{x}} + b)$. (W, b) is a parameter set of the encoder that can implement a mapping from the input data space to the feature space. W is a $n \times m$ weight matrix and b is a m dimensional offset vector.

Decoder: The decoder is a mapping $z = g(W'\overrightarrow{\mathrm{y}} + b')$ transforming the hidden representation $\overrightarrow{\mathrm{y}}$ back to the original representation $\overrightarrow{\mathrm{z}}$ with n dimensions in input data space. (W', b') is a parameter set of the decoder that can convert the feature space to the input data space based on a reconstruction function $g(\cdot)$.

We optimize W, b, W', b' by minimizing the reconstruction error $R(\overrightarrow{\mathrm{x}}, \overrightarrow{\mathrm{z}})$. Therefore, based on the optimal W, b, DAE can capture the crucial features of $\overrightarrow{\mathrm{x}}$.

Each layer is trained by minimizing the error in reconstructing its input. If all layers are trained, the network carries out a supervised fine-tune. DAE can learn robust and valuable features from the time series data.

3.4. Long short-term memory

LSTM is a extension version of recurrent neural network (RNN), which can deal with the exploding and vanishing gradient problems [41]. LSTM is an effective model to learn the features from time series data. Fig. 3 presents how information flows through a memory cell and is controlled by several gates. A common LSTM unit consists of four basic structures: a memory cell, an input gate, an output gate and a forget gate. It utilizes these four basic structures to keep the long term and short term dependencies in sequential data. The flow of information into and out from unit is controlled by three gates, and the value states are stored by the memory cell over arbitrary time intervals. The forget gate decides what information should be discarded from the memory cell, and this decision is controlled by the following formulas:

$$\mathrm{f_t} = \sigma\left(\omega_\mathrm{f}\overrightarrow{\mathrm{x}}_\mathrm{t} + \mathrm{U_f}\overrightarrow{\mathrm{h}}_{\mathrm{t-1}} + \mathrm{b_f}\right), \quad (4)$$

where $\mathrm{f_t}$ represents the forget gate; $\overrightarrow{\mathrm{x}}_t$ is the input vector at time t; $\omega_f \in \mathbb{R}^{N\times M}$, $\mathrm{U}_f \in \mathbb{R}^{N\times N}$, and $\mathrm{b_f} \in \mathbb{R}^{\mathrm{N}}$ represent the parameter matrices at all time steps with M and N denoting the dimension of input vectors and hidden states, respectively; $\overrightarrow{\mathrm{h}}_{t-1}$ denotes the previous output of the LSTM; σ is a sigmoid function $(\sigma(x) = 1/(1 + e^{-x}))$.

The input gate determines what information is updated. A tanh function creates a new candidate cell state $\tilde{C}_t$. Then update the cell state C_t by the following equations:

$$\mathrm{i}_t = \sigma\left(\omega_\mathrm{i}\overrightarrow{\mathrm{x}}_t + \mathrm{U_i}\overrightarrow{\mathrm{h}}_{\mathrm{t-1}} + \mathrm{b_i}\right), \quad (5)$$

$$\tilde{C}_t = \tanh\left(\omega_\mathrm{c}\overrightarrow{\mathrm{x}}_\mathrm{t} + \mathrm{U_c}\overrightarrow{\mathrm{h}}_{\mathrm{t-1}} + \mathrm{b_c}\right), \quad (6)$$

$$C_t = f_t \odot C_{t-1} + \mathrm{f_t} \odot \tilde{C}_t, \quad (7)$$

where $\mathrm{i_t}$ is the input gate; $\odot$ denotes the element wise product; ω_i and $\mathrm{b_i}$ are the parameter matrices of input gate; ω_c and $\mathrm{b_c}$ are the parameter matrices of the candidate cell; tanh is the hyperbolic tangent function $(\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x}))$.

The output gate O_t decides what information of the memory cell is sent out. The output gate is controlled by the following equation:

$$O_t = \sigma\left(\omega_\mathrm{o}\overrightarrow{\mathrm{x}}_\mathrm{t} + \mathrm{U_o}\overrightarrow{\mathrm{h}}_{\mathrm{t-1}} + \mathrm{b_o}\right), \quad (8)$$

where ω_o and $\mathrm{b_o}$ are the parameter matrices of the output gate.

The hidden state $\overrightarrow{\mathrm{h}}_\mathrm{t}$ at time t is obtained by

$$\overrightarrow{\mathrm{h}}_\mathrm{t} = O_t \odot \tanh(C_t). \quad (9)$$

What differentiates LSTM from the traditional neural network is its loop construct is carried out basically following a chronological order. Fig. 4 shows the architecture diagrams of LSTM. From Fig. 4, the input data is a vector at time t, and the status of hidden layer can transmit backward in accordance with the order of time. Therefore, a sequence of eigenvectors is required as a input data of LSTM, and the sequence is composed of eigenvectors of m consecutive time steps. Before training, the input sequence of LSTM should be constructed. Let $\overrightarrow{\mathrm{x}}_t$ be the eigenvector at time t, the sequential form is $\{x_{t-M+1}, x_{t-M+2}, \ldots, x_t\}$. Hence, the first sequence is $\{x_1, x_2, \ldots, x_M\}$, the second sequence is $\{x_2, x_3, \ldots, x_{M+1}\}$, and so on. Fig. 5 presents the structural diagram of the input sequence in LSTM.

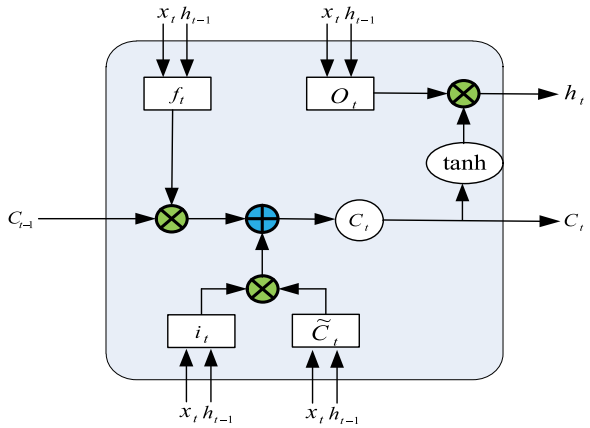


Fig. 3. An illustrative graph of the LSTM unit.

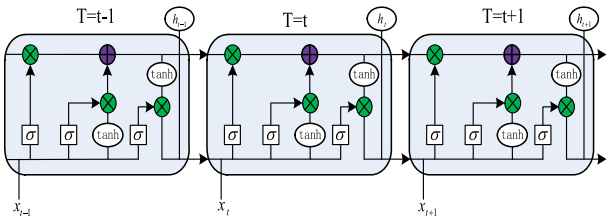


Fig. 4. Architecture of LSTM.

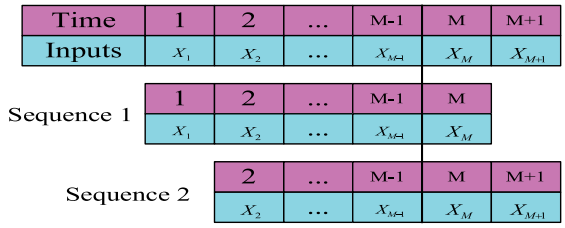


Fig. 5. The structural diagram of the input sequence in LSTM.

3.5. The proposed model

The architecture of SA-DLSTM is presented in Fig. 6. This model includes three parts. The first part is the deep neural network, which uses DAE models to extract the features from the original input vectors. The second part is the SA model which can obtain the sentiment index from the textual contents on Internet. The third part is the LSTM model used to predict stock market tendency. The last layer representation of DAE models which presents the generalization and representation of financial market environment is considered as the input for LSTM. The sentiment index which is the output of SA is fed into LSTM as the emotional representation. Two sources of data are collected and incorporated into LSTM: sentiment index and historical financial time series. Particularly, SA-DLSTM captures the intrinsic relationships among different information sources to improve the performance.

SA-DLSTM utilizes the historical financial quantitative data, news and social media data to forecast the stock market movement direction. Firstly, sentiment information is extracted from the textual contents by ECNN. Secondly, DAE models can extract features from financial quantitative data and technical indicators. Lastly, LSTM aggregates the features from ECNN and DAE models, and make stock market prediction. Algorithm 1 describes the training process of SA-DLSTM. The vector $\overrightarrow{\mathrm{x}}$ and T are the inputs of the Algorithm 1. The vector $\overrightarrow{\mathrm{x}}$ denotes stock price each day. It includes 6 features (the opening prices, the highest price, the lowest price, the closing price, the trading volume, and change amount) for all trading days, which is the same to that of stock market comment we have crawled. Firstly, let $\overrightarrow{\mathrm{x}}$ be the input of DAE models. Then DAE gets the crucial features $\overrightarrow{\mathrm{g}}$ of the financial data. Therefore, m vectors form a $m \times 6$ matrix as the input of DAE. m is the total trading days. The T is the financial textual comments. Let T be the input of ECNN. The output of ECNN is the sentiment index S_t at day t. Use Eq. (3) to revise the sentiment index S_t, and get the new sentiment index S_t^{rev}. In this paper, each comment is turned into a word vector matrix ($n \times 128$), where n is the total words of one comment, and 128 is the number of each word vector.

Algorithm 1: The training process of SA-DLSTM

Input: The vector $\overrightarrow{\mathrm{x}}$ denotes stock price; T is the financial textual comments; D denotes the days of the financial samples; Let e present the maximum number of iterations.
Output: the predicted result of the stock price.

```
Let T be the input of ECNN ;
The output of ECNN is the sentiment index S_t at day t;
Use Eq. (3) to revise the sentiment index S_t , and get the
  new sentiment index S_t^rev;
Let x⃗ be the input of DAE models ;
DAE models obtain the crucial features g⃗ of the financial
  data ;
Combining S_t^rev with g⃗, the new input of LSTM F⃗ is
  formed ;
for i = 1 : e do
    for t = 1 : D do
        Use Eq. (4)–(9) to update memory cell, forget gate,
          input gate, output gate ;
Obtain the predicted result ;
```

4. Experimental validation

To evaluate effectiveness of SA-DLSTM, we use both stock forum data and financial data to test the performance of SA-DLSTM. Models are trained and evaluated by accuracy on HSI, and SA-DLSTM is used for market simulation by using prediction results as trading signals. The HSI is the most widely quoted barometer for the Hong Kong economy and is often used as a market benchmark for Hong Kong investors.

4.1. Data description

The HSI is a market capitalization-weighted index of the largest companies that trade on the Hong Kong Exchange. It reflects the performance of many leading and influential enterprises in Hong Kong securities market. The stock data includes opening price, closing price, high price for the day, low price for the day, and trading volume. In this paper, we select the daily trading data of HSI from January 2002 to December 2020. Meanwhile, we use the web crawler to download the textual contents of the HSI from the Sina stock form, Eastmony stock forum and China Microblog. In this paper, we select the 100 most discussed comments each day to obtain the sentiment indexes. An financial textual comment example is “American index and European index climbed higher yesterday, so did Hong Kong and Japan index, and China A-share index up a little :)”, where :) is the smiling face.

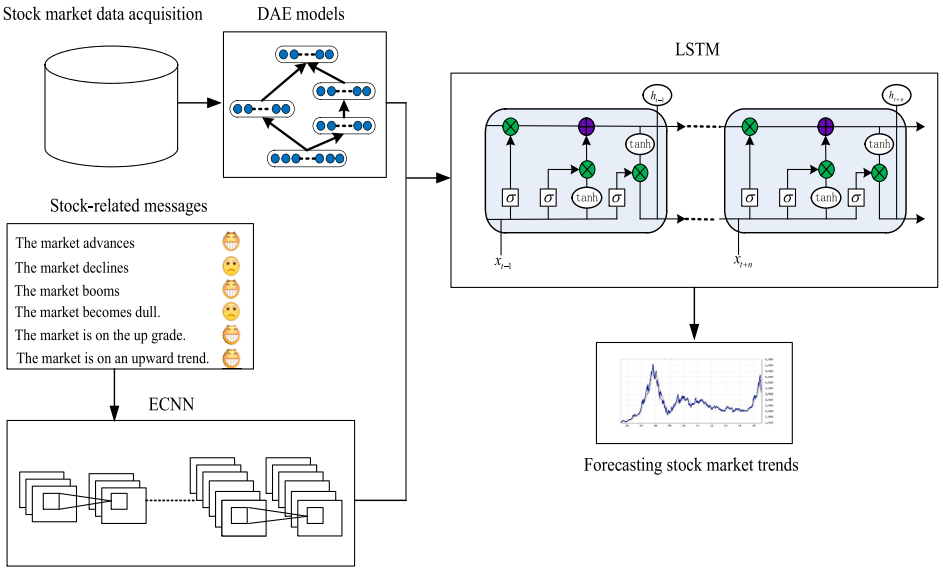


Fig. 6. Framework of SA-DLSTM.

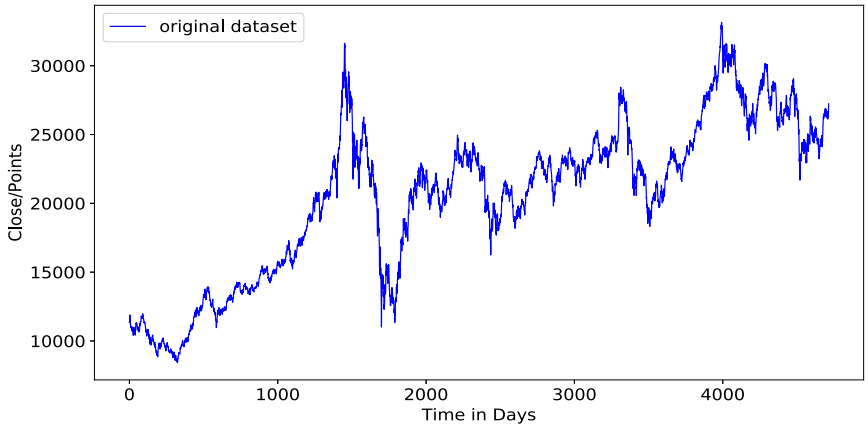


Fig. 7. The closing price of HSI.

It is generally known that the closing price is the price recognized by all market participants. It contains a lot of useful information to predict the price trend of the next trading day. Therefore, we uses the closing price of HSI on the next trading day as the prediction object. Fig. 7 presents the closing price of HSI over the period between 2002 and 2020. From Fig. 7, HSI initially shows an upward trend, followed to descend, then takes on ascend trend, and finally presents an oscillation trend.

In order to improve the generalization ability of SA-DLSTM and avoid over-fitting problems, we use the data for the past 3 years to construct the input sequence and as the training dataset to train our model. The data of the next 3 months are used to construct the test dataset, and the trained model is used to predict the closing price of the test dataset. For example, we use the data from January 2002 to December 2004 as the training dataset, and the data from January 2005 to March 2005 as the test dataset. Then the training dataset and test dataset are pushed back 3 months respectively, and the process is repeated until all samples are covered. In each round of training, we split the training dataset into a 80%–20% ratio to create the validation dataset. The process is shown in Fig. 8.

4.2. Evaluation metrics

In order to evaluate the performance of SA-DLSTM, we use the Mean Absolute Percentage Error (MAPE) to quantify the performance. MAPE measures the relative deviation between the true and the predicted values. The smaller MAPE is, the closer the predicted value is to the real value. The MAPE is defined as follows:

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{c_i - \widetilde{c_i}}{c_i}\right| \times 100\%, \tag{10}$$

where n is the number of all samples, c_i and $\widetilde{c_i}$ are the true and predicted values of the ith sample, respectively.

4.3. Result discussion

In order to improve the prediction accuracy of SA-DLSTM, the input is preprocessed. We transform the original features into a number in [0, 1] before extracting features to eliminate influence of dimension. The linear transformation formula is define as

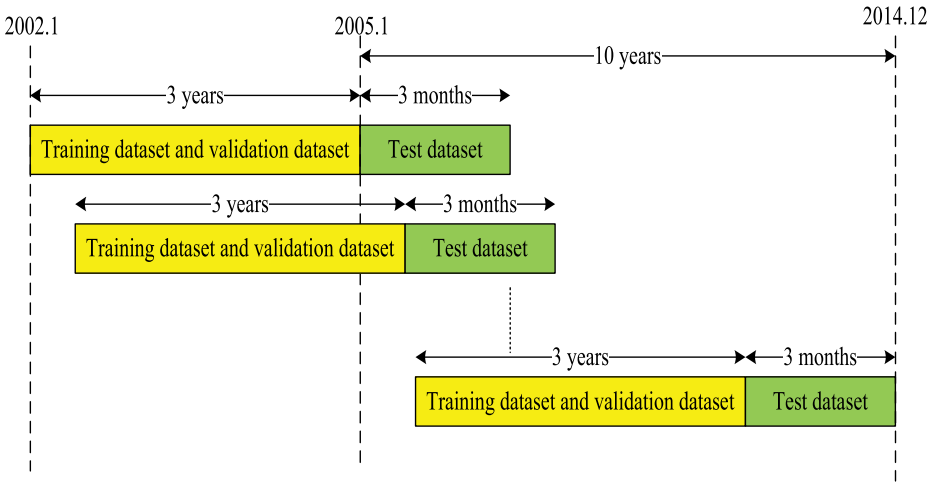


Fig. 8. The diagram of dataset partition.

Table 1
MAPE of different methods.

Year	WPCA-NN	WNN	LSTM	SVM	LR	LS_RF	SA-DLSTM
2005	3.2	3.6	1.6	2.9	4.7	2.7	**0.9**
2006	3.5	5.5	2.1	4.8	6.1	3.1	**0.8**
2007	10.8	50.6	5.7	41.4	53.3	8.3	**1.2**
2008	3.9	40.9	2.1	39.1	43.2	3.3	**2.3**
2009	3.6	11.2	3.1	11.9	12.2	2.9	**1.4**
2010	3.4	8.5	2.4	7.8	9.4	3.2	**1.3**
2011	3.6	15.8	2.6	12.9	17.7	4.1	**1.2**
2012	2.9	5.5	1.6	4.9	5.9	2.2	**0.8**
2013	2.4	5.2	1.7	5.1	6.1	2.9	**0.7**
2014	3.6	5.3	1.5	4.7	5.7	2.7	**0.9**
2015	3.3	5.4	1.3	4.6	5.8	1.8	**0.8**
2016	3.7	5.1	1.6	5.0	5.4	3.1	**0.9**
2017	4.3	7.8	2.7	5.9	6.7	4.2	**1.1**
2018	3.8	6.2	2.1	4.8	5.6	2.9	**1.3**
2019	2.8	4.8	1.6	3.9	4.8	2.1	**0.6**
2020	3.8	5.7	2.2	4.2	6.3	3.3	**1.4**
Average	3.91	11.69	2.24	10.24	12.43	3.3	**1.10**

follows:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}. \tag{11}$$

In the process of features extraction by DAE models, the Gaussian noise is added to the original data. We use the sigmoid function as the activation function in DAE. The structure of SA-DLSTM is ascertained after repeated experiments. DAE models integrates both single layer and multi-layers DAEs. The single layer DAE has the number of neurons is 8. The multi-layers DAE has 4 layers, and the numbers of neurons on each layer are 16,8,8, and 8, respectively. The number of LSTM is 2, and the numbers of hidden layer neurons for LSTM are 8 and 8, respectively. The training timestep is 5, that is, the length of the input sequence is 5. The batch size of LSTM is 50, and the epoch of LSTM is 1000.

In order to evaluate performance of SA-DLSTM, traditional machine learning methods and statistical models are used as baselines. Specifically, we build single type networks using convolutional unit and LSTM unit as comparisons to test the performance of SA-DLSTM. In this section, we use the Wavelet-PCA neural network (WPCA-NN) [42], wavelet neural network (WNN) [43], LSTM [44], SVM [23], Linear regression (LR) and LS_RF [45] methods to compare with SA-DLSTM. Table 1 shows the comparison results of MAPE of all methods on testing dataset. It can be seen from the Table 1 that the MAPE of SA-DLSTM is better than other models for closing price prediction. The average MAPE of LSTM is 2.37%, while that of SA-DLSTM is 1.12%. LSTM performs worst, and the possible reason it that it just depends on the financial data for prediction and cannot consider the stock-related textural contents which have been proven to be effective for stock prediction. Fig. 9 shows the prediction results under each model. As can be seen from Fig. 9, the predicted closing price of SA-DLSTM is the closest to the actual value. The experimental result shows that the sentiment index can improve performance. Fig. 9 and Table 1 can conclude that (1) the integration model of multiple models can enhance the efficiency of feature extraction since SA-DLSTM has a better performance than other single model methods; (2) the models which use deep learning methods have better prediction performance since LTSM and SA-DLSTM outperform other traditional methods.

In Fig. 9, compared to LSTM, SA-DLSTM is closer to actual value, indicating that the sentiment can improve the prediction accuracy. And DAE models can also improve SA-DLSTM's performance. In financial market, not each participant acts in a rational way based on the theoretical model. Human un-rational actions play an important role in the financial market. Based on human psychological characteristics and behavioral characteristic, the behavioral finance finds that stock market changes are related to the psychological and behavioral characteristics of participants.

To verify whether the prediction accuracy of SA-DLSTM was significantly better than that of other methods, the Friedman test [46] was applied to investigate whether the hit ratio achieved by SA-DLSTM and other methods were significantly different. Table 2 presents the Friedman test results, from which we find that the conclusion is significant at the 0.05 level for one-tailed

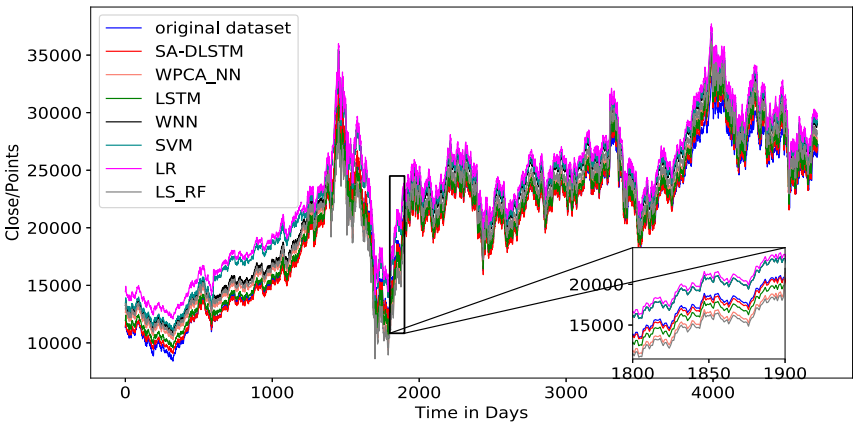


Fig. 9. The predicted results of different methods.

Table 2
Friedman test results for SA-DLSTM against other methods.

Compared models	Significant level $\alpha = 0.05$
SA-DLSTM vs. WPCA-NN	H0: n1=n2=n3=n4=n5=n6
SA-DLSTM vs. WNN	F = 12.38
SA-DLSTM vs. LSTM	p = 0.05 (reject H0)
SA-DLSTM vs. SVM	
SA-DLSTM vs. LR	
SA-DLSTM vs. LS_RF	

test. It demonstrates that SA-DLSTM significantly outperformed other methods.

The improvement of SA-DLSTM comes from two aspects: (1) SA-DLSTM considers the influence of sentiment on stocks based on the behavioral finance; (2) Based on DAE models, SA-DLSTM obtains the key features from the original data, and pays attention to important information in large amounts of data. In this paper, we believe that the investors' sentiment has a significant impact on stock price.

4.4. Market simulation

To evaluate the effectiveness of SA-DLSTM on market prediction, this section simulates a prediction-based trading to test whether SA-DLSTM can get profit. In the simulation, a investor makes a trading decision according to predictions of SA-DLSTM based on a simple strategy: if SA-DLSTM predicts that the stock price has a rising trend, the system will buy this stock; if SA-DLSTM predicts that the stock price has a falling trend, the system will have a short position of the stock. In this paper, we assume that transaction cost is zero. In order to simulate this trading strategy, we use SA-DLSTM trained on training dataset to predict future trends of HSI and send trading signals.

MAPE can only evaluate the capability of the classification-based prediction, which corresponds to the future return. The profitability that is correlated to the amount of rise or fall can actually matter in the market practice. In order to evaluate effectiveness of SA-DLSTM in this section, the total return R can be expressed as:

$$R = \left(\frac{Porfit_T}{Porfit_{t_0}} - 1\right) \times 100\%, \tag{12}$$

where $Porfit_T$ is the assets in the account at the end of simulation period T, $Porfit_{t_0}$ is the assets in the account at the start of simulation period t_0. The investment income is higher, possibly the profitability performance of the model is better. Generally, the annual return AR can be measured as:

$$AR = (1 + R)^{247/T_s} - 1, \tag{13}$$

where T_s is the time span (days) during market simulation, where T_s is the time span (days) over which SA-DLSTM is simulated, 247 is the average trading days for a year. The risk-adjusted profit is a key indicator which is applied to evaluate the stability of a trading system, hence the Sharp ratio SR has been utilized in many trading related studies [47]. The SR is defined as:

$$SR = \sqrt{247} \times \frac{\overline{r_e}}{\sigma_e}, \tag{14}$$

where

$$\overline{r_e} = \frac{1}{n}\sum_{i=1}^{n}\left[r_p(i) - r_f(i)\right], \tag{15}$$

$$\sigma_e = \sqrt{\frac{1}{b-1}\sum_{i=1}^{n}\left[r_p(i) - r_f(i) - \overline{r_e}\right]^2}, \tag{16}$$

where $\overline{r_e}$ is the average daily excess returns, σ_e is the volatility of the daily excess returns, $r_p(i)$ is the returns from the trading strategy, $r_f(i)$ is the risk-free rate of interest on the ith trading day, n is the number of trading days. A larger SR implies a high profit. To evaluate the risk of the trading strategy, we use the annual volatility V which is measured as:

$$V = \sqrt{\frac{247}{n-1}\sum_{i=1}^{n}\left[r_p(i) - \overline{r_p}\right]^2}, \tag{17}$$

where $\overline{r_p}$ is the average return from the trading strategy.

Table 3 shows the results of each method in market simulations. From Table 3, we can find that SA-DLSTM can capture suitable trading points to make profits. Among these methods, methods based on machine learning and deep learning models have better returns and annual returns than LR. It indicates that the traditional statistical models show less profitability than the non-linear ones. From Table 3, SA-DLSTM outperforms other methods at returns. The LR still shows a better performance in terms of the risk control, and the annual volatility (1.83%) is only one-quarter of that of SA-DLSTM (7.32%). This implies that LR has less risky than other methods based on machine learning models for it is a conservative strategy that leads to less returns.

In the view of stability, the most stable result from SA-DLSTM reaches 4.43 sharpe ratio. It can be found that based on the integrated structure, the units that are integrated can affect the quality of capturing signals. It has better capability of capturing profitable and stable signals than other methods.

Table 3
Market simulation results.

	R	AR	SR	V
WPCA-NN	15.27%	19.71%	1.72	8.96%
WNN	12.86%	16.74%	1.43	9.89%
LSTM	19.81%	23.17%	1.86	9.36%
SVM	13.87%	17.36%	1.67	8.69%
LR	8.75%	10.16%	3.32	**1.83%**
LS_RF	18.29%	21.76%	1.72	9.81%
SA-DLSTM	**27.63%**	**41.27**	**4.43**	7.32%

5. Conclusion

In this paper, a novel LSTM-based model with sentiment is proposed to predict stock market trend. Investor psychology is an important role to drive the stock market. Therefore, we use the sentiment index to take the investor's emotional into consideration. We can find a lot of textual content to reflect investor psychology from the Internet. This paper uses SA to convert textual documents into daily sentiment indexes. The LSTM-based model is improved by DAE models, which extract the key features from the complex financial data. DAE models can help the model focus on the most contribution information of the current task.

According to the experiments, the performance of SA-DLSTM has been verified. The empirical results show that SA-DLSTM outperform other methods. The experimental results validate that combining sentiment index with stock market data can improve performance to predict stock market trend. Therefore, sentiment that contains valuable information about the basic value of financial assets is regarded as one of the key indicators of the stock market. For future work, it is imperative to improve the efficiency of SA-DLSTM to process voluminous data in real time.

CRediT authorship contribution statement

Yanli Zhao: Conceptualization, Methodology, Formal analysis, Investigation, Resources. **Guang Yang:** Software, Data curation, Writing – original draft, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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